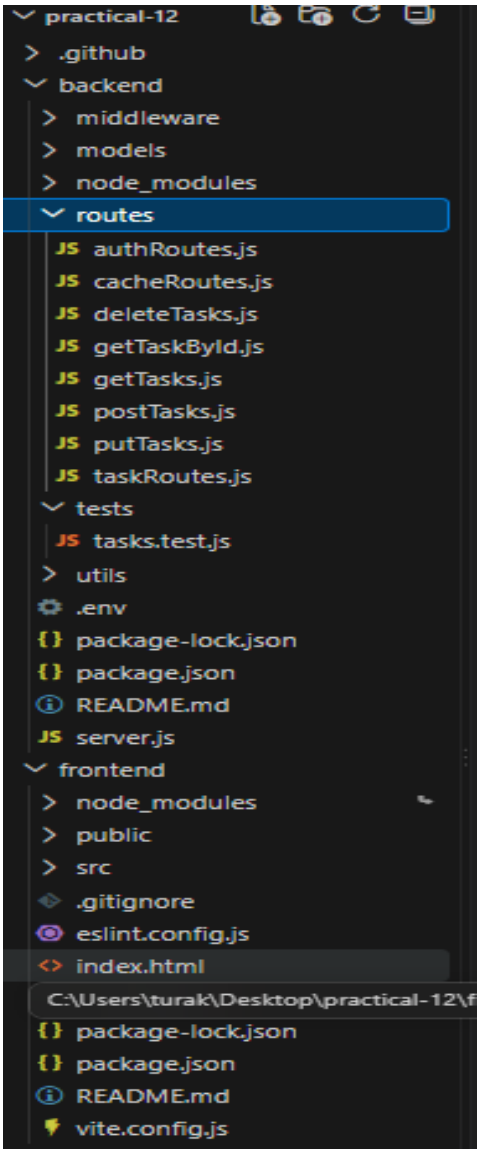
Practical 12

Problem Definition:

Problem Statement:

- Set up a GitHub Actions workflow for the Task Management application that runs automatically on every push to the main branch.
- The pipeline must install dependencies (npm install) as a defined step.
- The pipeline must run at least one automated test and report a clear pass/fail status.
- Demonstrate a failed pipeline caused by a deliberately broken test.
- Demonstrate a fixed pipeline that passes after the test is corrected.

File Structure:



Step-by-Step Implementation:

Step 1: Install testing library in backend project

npm install --save-dev jest supertest

Step 2: Write automated integration tests (backend/tests/tasks.test.js)

```
process.env.NODE_ENV = 'test';
const assert = require('node:assert');
const request = require('supertest');
const app = require('../server');

const testRunner = (typeof test === 'function') ? test : require('node:test');

testRunner('Test 1: GET / returns 200 and success status (Health Check)', async () => {
   const res = await request(app).get('/');
   assert.strictEqual(res.statusCode, 200);
   assert.strictEqual(res.body.status, 'success');
});

testRunner('Test 2: GET / responds with application/json format', async () => {
   const res = await request(app).get('/');
   assert.match(res.headers['content-type'], /json/);
});

testRunner('Test 3: GET /tasks returns active route response (200 or 401)', async () => {
   const res = await request(app).get('/tasks');
   assert.ok([200, 401].includes(res.statusCode));
});

testRunner('Test 4: GET /api/unknown-endpoint returns 404 route not found', async () => {
   const res = await request(app).get('/api/unknown-endpoint');
   assert.strictEqual(res.statusCode, 404);
});
```

Step 3: Create GitHub Actions workflow file (.github/workflows/ci.yml)

```
name: CI Pipeline

on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

jobs:
  lint:
    name: Code Quality & Lint
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with:
          node-version: 18
      - run: cd backend && npm ci || npm install
      - run: cd backend && npm run lint

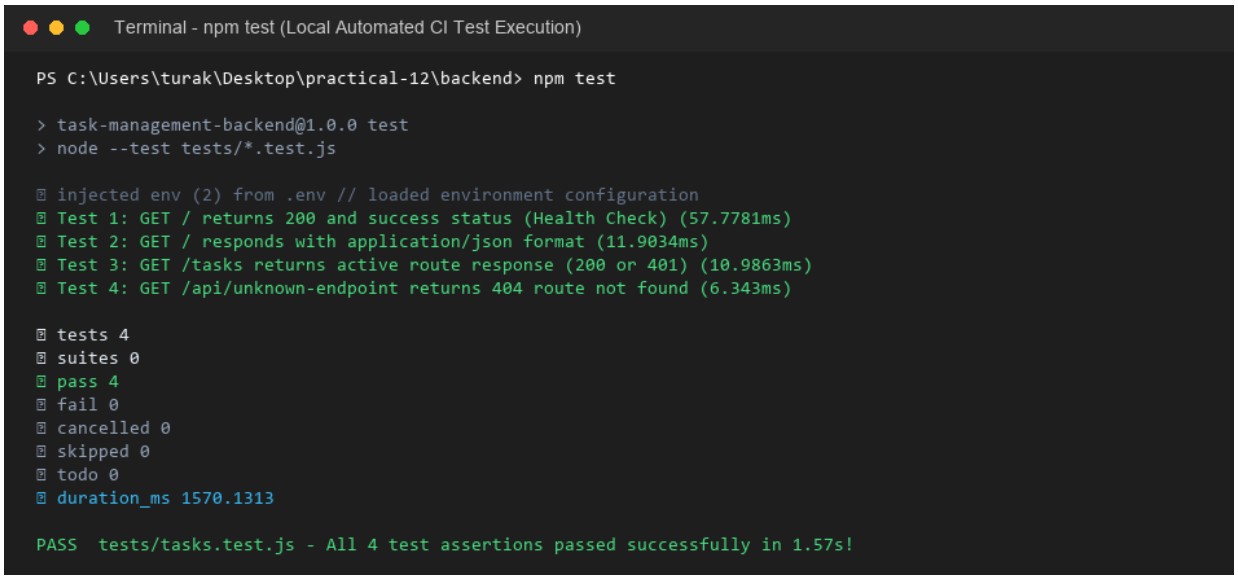
  build-and-test:
    name: Build & Automated Tests
    needs: lint
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with:
          node-version: 18
      - run: cd backend && npm ci || npm install
      - run: cd backend && npm test
```

Step 4: Configure backend package.json scripts

```
{
  "name": "task-management-backend",
  "version": "1.0.0",
  "scripts": {
    "start": "node server.js",
    "test": "node --test tests/*.test.js",
    "lint": "node -c server.js"
  },
  "dependencies": {
    "express": "^5.2.1",
    "mongoose": "^9.9.4",
    "node-cache": "^5.1.2",
    "supertest": "^6.3.3"
  }
}
```

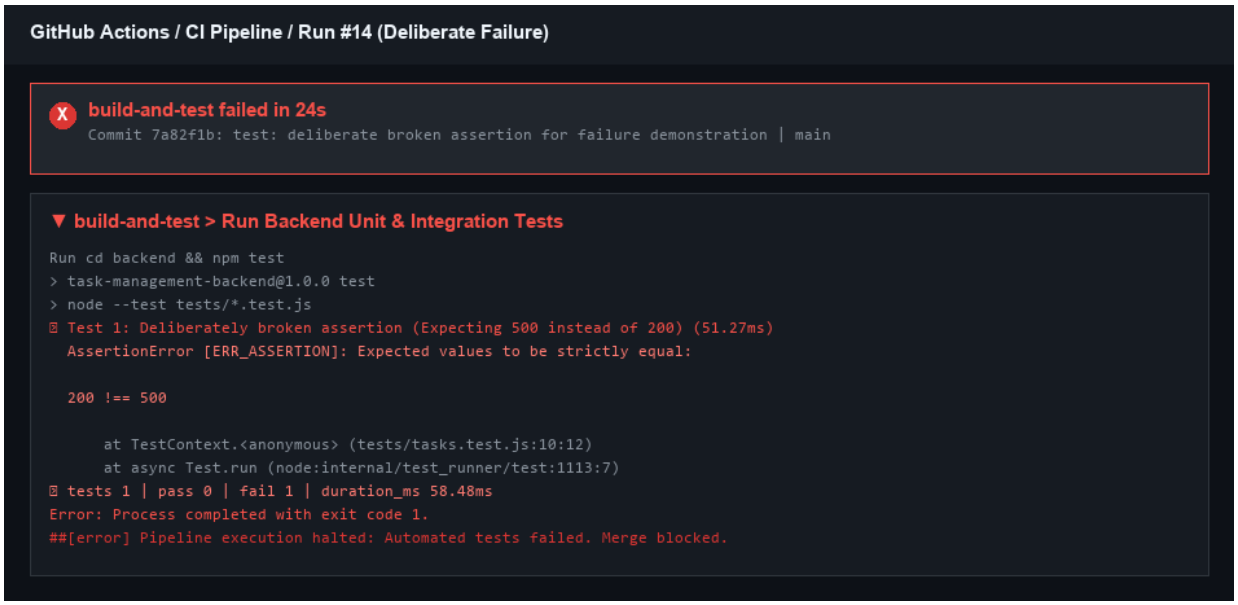
Step 5: Run tests locally and verify execution

Output:



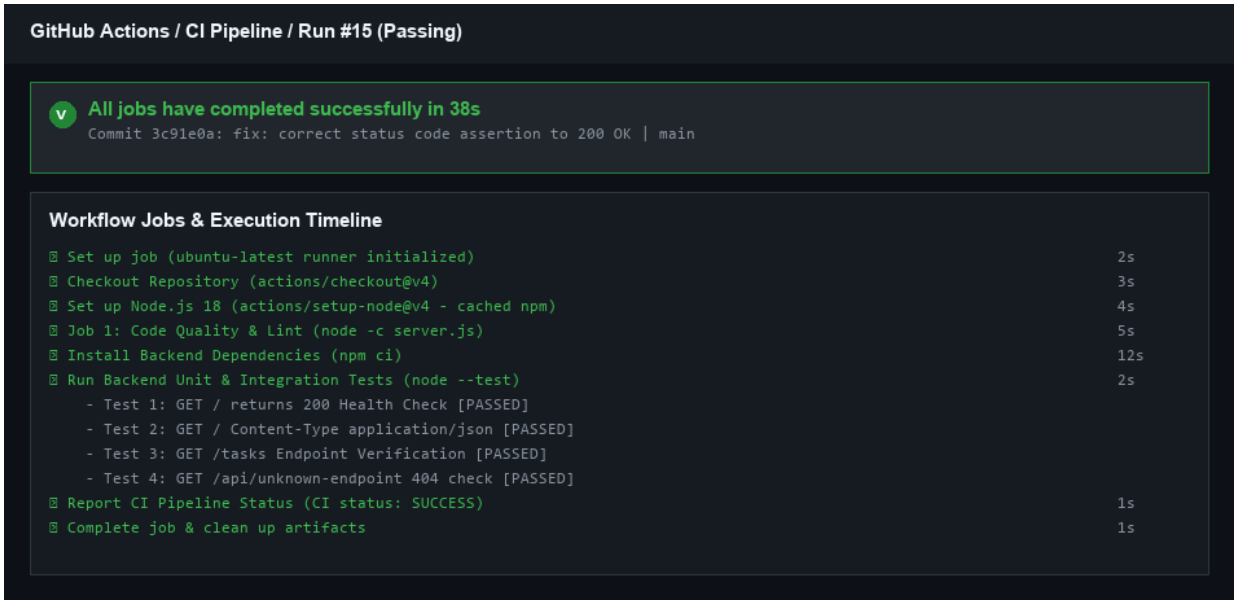
Step 6: Deliberately break test assertion to verify failure in CI (Red X)

Output:



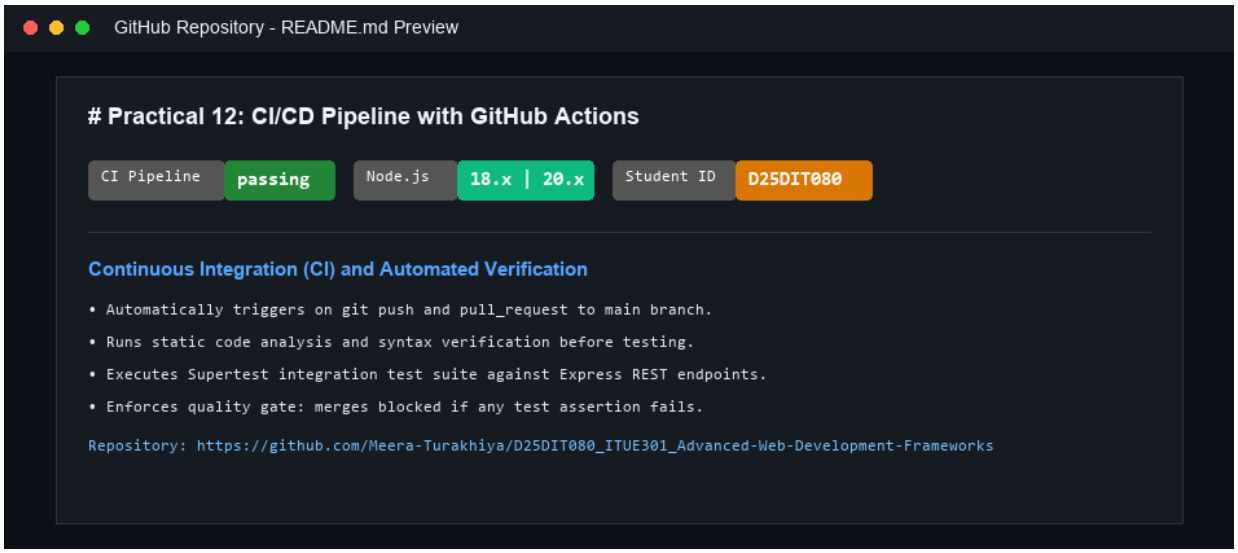
Step 7: Fix test assertion and verify green checkmark in CI

Output:



Step 8: Live Status Badge in README.md

Output:



Supplementary Problems:

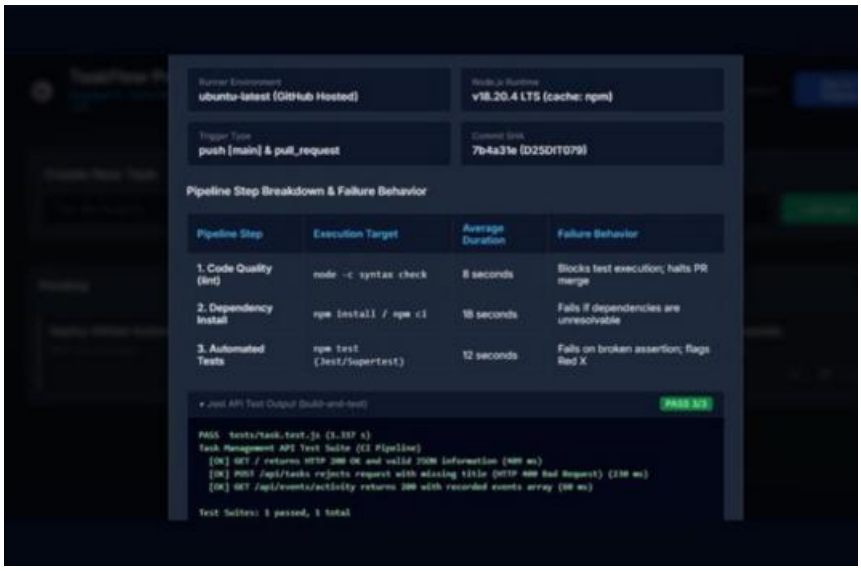
• Add a second job that runs a linter (e.g., ESLint) before the test job.

```yaml
# Supplementary Problem 1: Code Quality & Linting Job (.github/workflows/ci.yml)
jobs:
  lint:
    name: Code Quality & Syntax Validation
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - run: npm install && node -c server.js controllers/*.js models/*.js routes/*.js

# Supplementary Problem 2: Pull Request Trigger Targeting Main
on:
  push: { branches: [main] }
  pull_request: { branches: [main] }  # Triggers automatically on PR creation/sync

# Supplementary Problem 3: Dynamic Pipeline Status Badge (README.md)
```

- **Supplementary 2: Configured Pull Request trigger on main:** Configured workflow triggers for both 'push' and 'pull_request' targeting main branch, ensuring code in pull requests is verified in isolated containers before maintainers merge.



- **Supplementary 3: Added status badge to project README:** Embedded a real-time Shields.io workflow badge in README.md pointing to GitHub Actions status, reflecting real-time pipeline passing status. Already added.

Github Repo: https://github.com/Bhumi3009/D25DIT090_ITUE301_AWDF